

Tables from BigQuery Database

1. Citibike Stations table
https://console.cloud.google.com/bigquery?ws=!1m5!1m4!4m3!1sbigquery-public-data!2snew_york_citibike!3citibike_stations
2. Citibike trips table
https://console.cloud.google.com/bigquery?ws=!1m5!1m4!4m3!1sbigquery-public-data!2snew_york_citibike!3citibike_trips

Note: There are 4 pages in total

Subquery SELECT

The screenshot shows the Google Cloud BigQuery console interface. At the top, there are tabs for the query editor and the results. The query editor tab is active, showing a SQL query. Below the query editor, there is a status bar indicating the query is completed and the processing quota used. The results tab is also visible, showing a table with 6 rows and 4 columns: Row, station_id, num_bikes_available, and avg_num_bikes_available. The table data is as follows:

Row	station_id	num_bikes_available	avg_num_bikes_available
1	495	0	12.39186991869...
2	3171	0	12.39186991869...
3	3603	0	12.39186991869...
4	3733	0	12.39186991869...
5	3792	0	12.39186991869...
6	4170	0	12.39186991869...

Subquery FROM

```
SELECT
-- 1. Grab the official Station ID from the 'stations' table
station_id,

-- 2. Grab the human-readable Name from the 'stations' table
name,

-- 3. Grab the calculated count from our subquery below and rename it for the final report
number_of_rides AS number_of_rides_starting_at_station

FROM
-- 4. OPEN SUBQUERY: We need to build a temporary "summary table" of ride counts first
(
  SELECT
    -- 5. Convert the ID from Number to Text so it can "talk" to the other table later
    CAST(start_station_id AS STRING) AS start_station_id_str,

    -- 6. Count the total rows in each group (this calculates total rides per station)
    COUNT(*) AS number_of_rides

  FROM
    -- 7. The source for the subquery: the massive log of every single bike ride
    bigquery-public-data.new_york.citibike_trips

    -- 8. Group the thousands of individual rides into piles based on their Station ID
    -- (Note: Must repeat the CAST here so SQL knows how to sort the piles correctly)
  GROUP BY
    CAST(start_station_id AS STRING)
)

-- 9. Name this temporary subquery table "station_num_trips" so we can refer to it
AS station_num_trips

-- 10. Connect our temporary summary table to the official Stations table
INNER JOIN
  bigquery-public-data.new_york.citibike_stations

-- 11. The "Handshake": Match rows where the Station ID (text) equals the Converted ID
(text)
ON
  station_id = start_station_id_str

-- 12. Sort the final list so the most popular stations appear at the top
ORDER BY
  number_of_rides DESC;
```

RESULT of the query

Untitled query

RunScheduleOpen inSaveDownloadShare

```
1 SELECT
2   -- 1. Grab the official Station ID from the 'stations' table
3   station_id,
4
5   -- 2. Grab the human-readable Name (e.g., "Central Park") from the 'stations' table
6   name,
7
8   -- 3. Grab the calculated count from our subquery below and rename it for the final report
9   number_of_rides AS number_of_rides_starting_at_station
10
```

Query completed

Using on-demand processing quota

Query results

Create conversationSave resultsOpen in

Job informationResultsVisualizationJSONExecution detailsExecution graph

Row	station_id	name	number_of_rides_...
1	497	E 17 St & Broadway	291615
2	293	Lafayette St & E 8 St	277060
3	435	W 21 St & 6 Ave	275348
4	426	West St & Chambers St	260911
5	285	Broadway & E 14 St	244420
6	151	Cleveland Pl & Sprina St	229694

Subquery WHERE

```
SELECT
  station_id, -- Select the unique ID of the station
  name        -- Select the human-readable name of the station
FROM
  bigquery-public-data.new_york.citibike_stations -- Source table: Station information
WHERE
  station_id IN -- Filter: Only keep stations that appear in the list below
  (
    SELECT
      -- Convert the trip's start station ID to text to match the outer table's format
      CAST(start_station_id AS STRING) AS start_station_id_str
    FROM
      bigquery-public-data.new_york.citibike_trips -- Source table: Trip history
    WHERE
      usertype = 'Subscriber' -- Filter: Only look at trips made by Subscribers
  );
```

RESULT of the query

The screenshot shows the Google Cloud BigQuery interface. At the top, there's a toolbar with buttons for 'Run', 'Schedule', 'Open in', 'Save', 'Download', and 'Share'. Below the toolbar, the query editor shows the SQL code from the previous block. A status bar indicates 'Query completed' and 'Using on-demand processing quota'. Below the query editor, the 'Query results' section is displayed, showing a table with 5 rows and 2 columns: 'station_id' and 'name'. The table is titled 'Query results' and has tabs for 'Job information', 'Results' (selected), 'Visualization', 'JSON', 'Execution details', and 'Execution graph'.

Row	station_id	name
1	3416	7 Ave & Park Pl
2	3397	Court St & Nelson St
3	419	Carlton Ave & Park Ave
4	3348	Coffey St & Conover St
5	3324	3 Ave & 14 St